Frugal Learning of Manipulation Skills

Swiss Robotics Day — November 13, 2026

Frugality in robot learning:

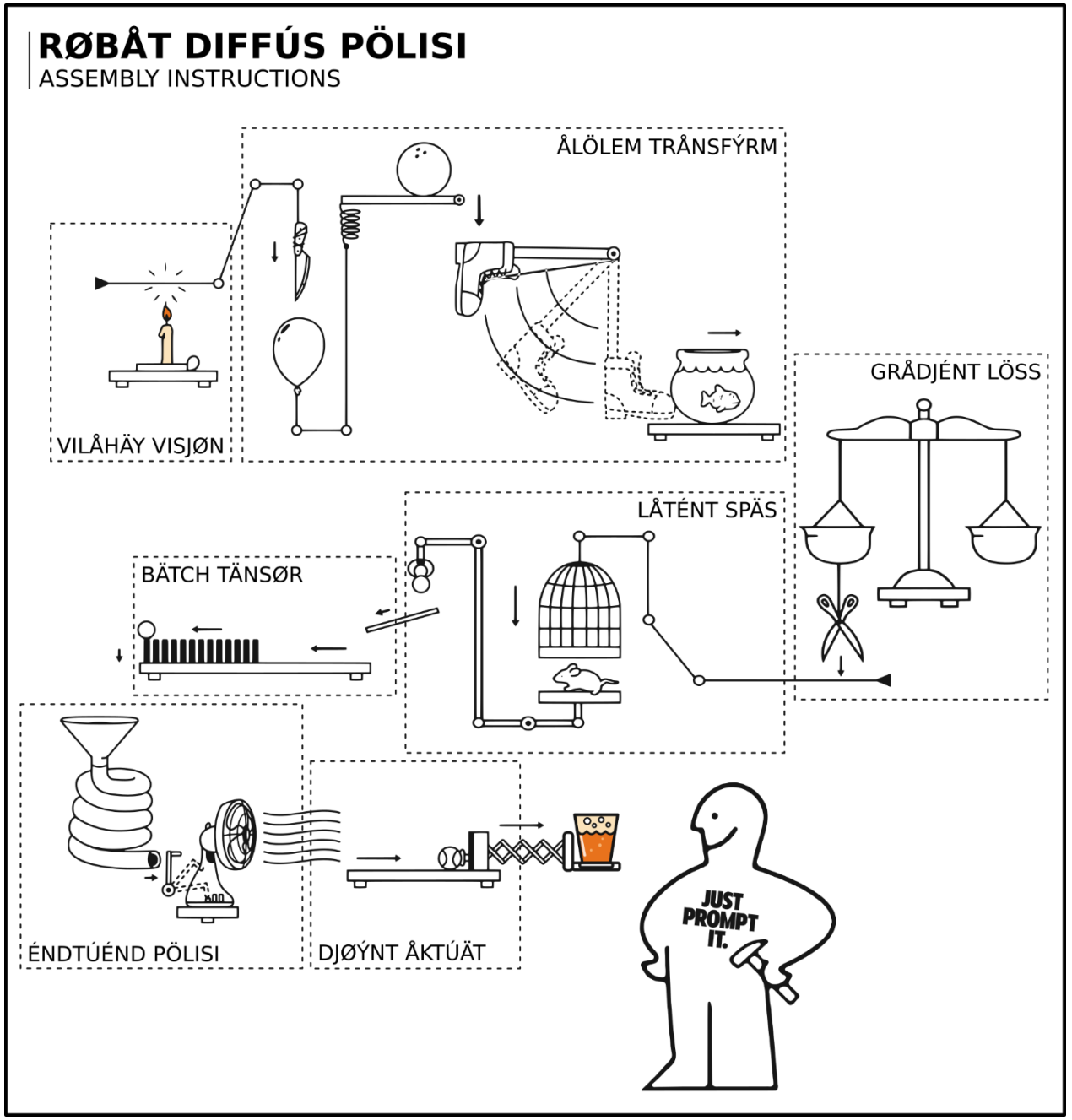
- in the amount of required data
- in what needs to be learned

Sylvain Calinon

https://calinon.ch

idiap RESEARCH INSTITUTE

AI for Society since 1991



Inspired by Joe Van Wetering, Lisa Wickmark, Jan Fredlund & Pär Zimmerdahl's artworks

Data Efficiency by Rethinking Search and Representation

Algorithms

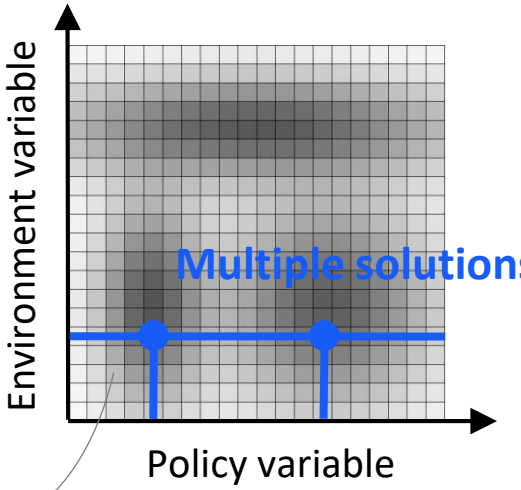
rethinking how to search and explore

Representations

rethinking how to model robots, objects and tasks

Adaptive skill learning and retrieval by low-rank factorization

Task parameters, current situations, objects locations, …



Given situations

Cost function redefined as a probability distribution

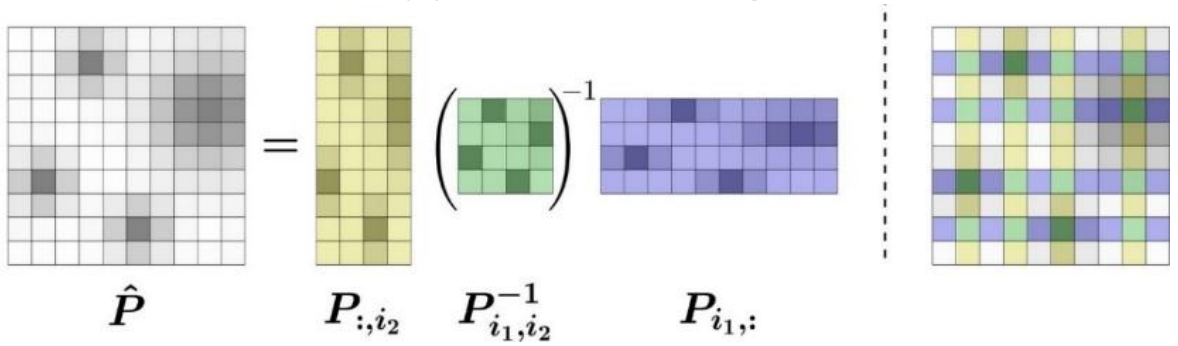
Actions, motions, controllers, behaviors, …

Adaptive skill acquisition = Estimating this distribution — 1-4 hour(s)

Skill generation = Conditional sampling — 1-10 millisecond(s)

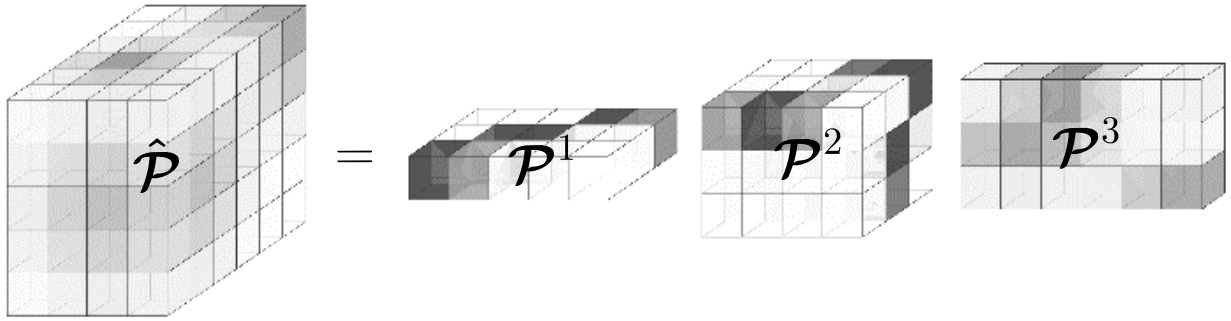
Skeleton decomposition of a matrix

(cross approximation algorithm)



for more dimensions

Tensor train decomposition

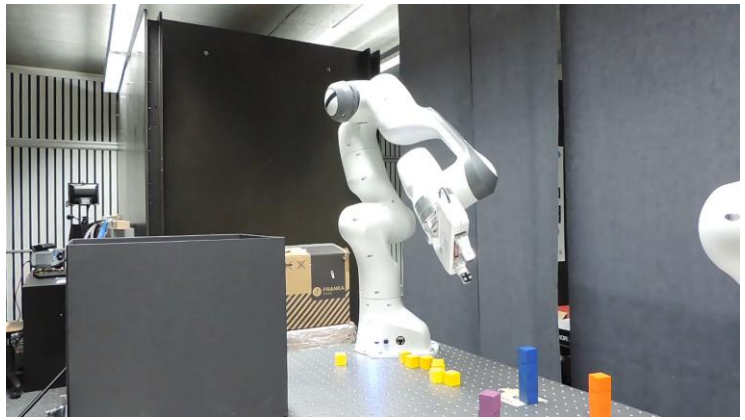


Our workhorse to make RL data-efficient:

Ivan V. Oseledets, "Tensor-Train Decomposition," SIAM Journal on Scientific Computing, 2011

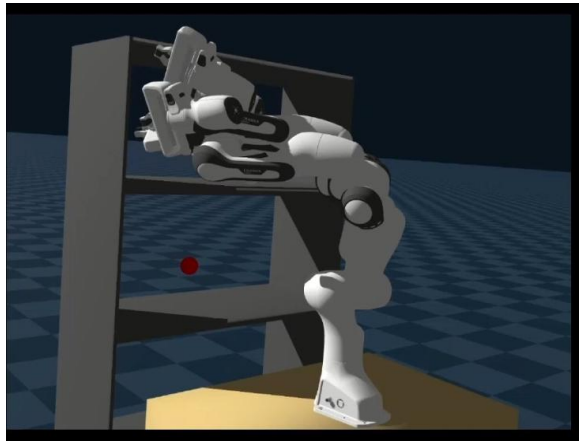
Tensor train (TT) as surrogate model

TT for global optimization (TTGO)



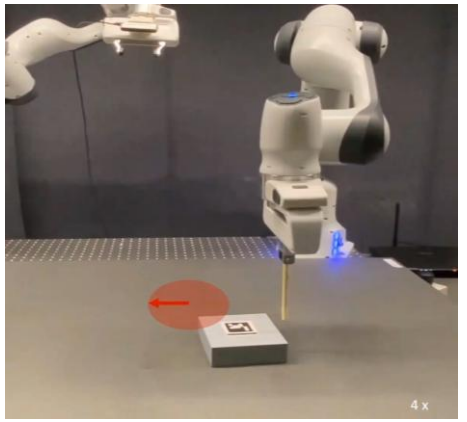
[Shetty, Lembono, Löw and Calinon, IJRR, 2024]
(Best article award, IEEE RAS TC OptRob)

Input:
Picking & placing locations
Output:
Full trajectories (multiple options)



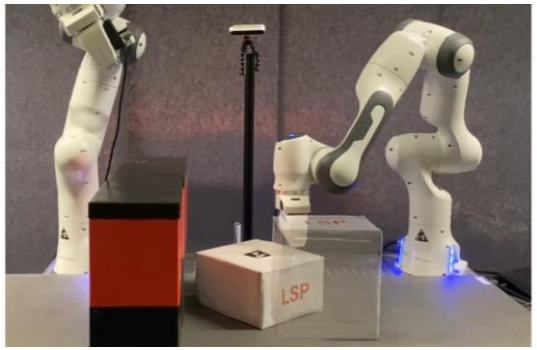
memmo
EC, H2020
(2018-2022)

TT for Dynamic Programming & RL



[Shetty, Xue and Calinon, ICLR 2024]
[Xue, Razmjoo, Shetty and Calinon, IJRR, 2025]

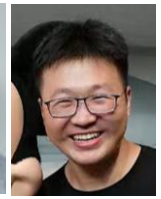
TT for constrained optimization & MPC



[Xue, Razmjoo, Shetty and Calinon, RSS 2024]
[Razmjoo, Xue, Shetty and Calinon, IJRR, 2025]


Dr Suhan Shetty


Dr Teguh Lembono

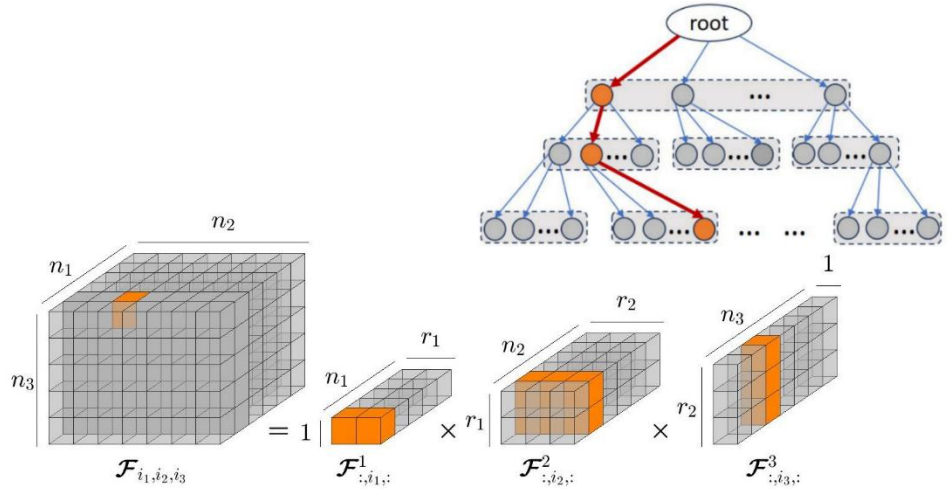

Dr Teng Xue


Dr Tobias Löw


Dr Amirreza Razmjoo

TT for Monte Carlo Tree Search (MCTS)

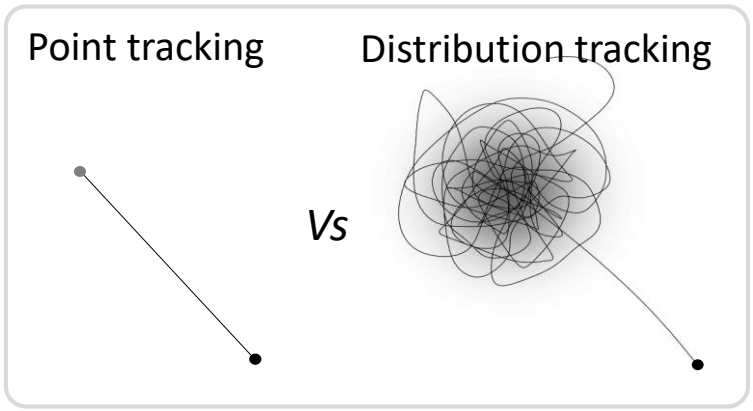
→ Low-rank factorization for tree search



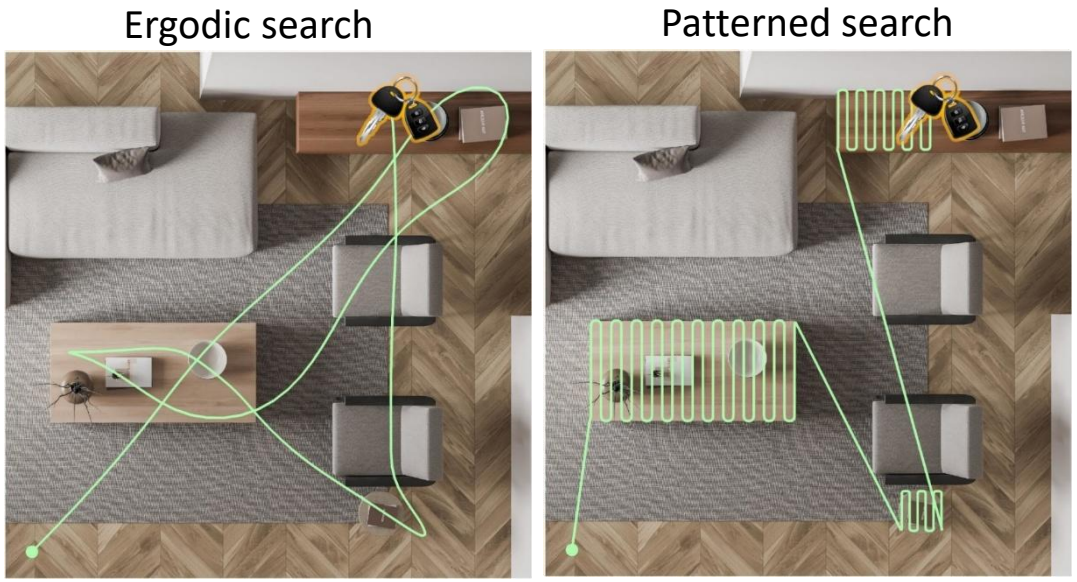
[Xue, Razmjoo, Zhang and Calinon, arXiv:2507.04949, 2026]

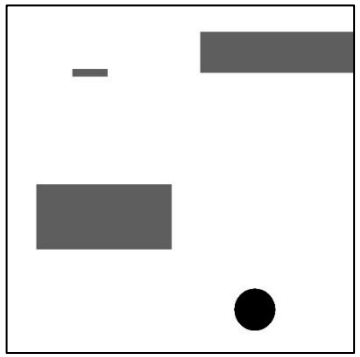
Ergodic control for search and exploration

Horizon Europe (2022-2025)
Horizon Europe (2022-2026)

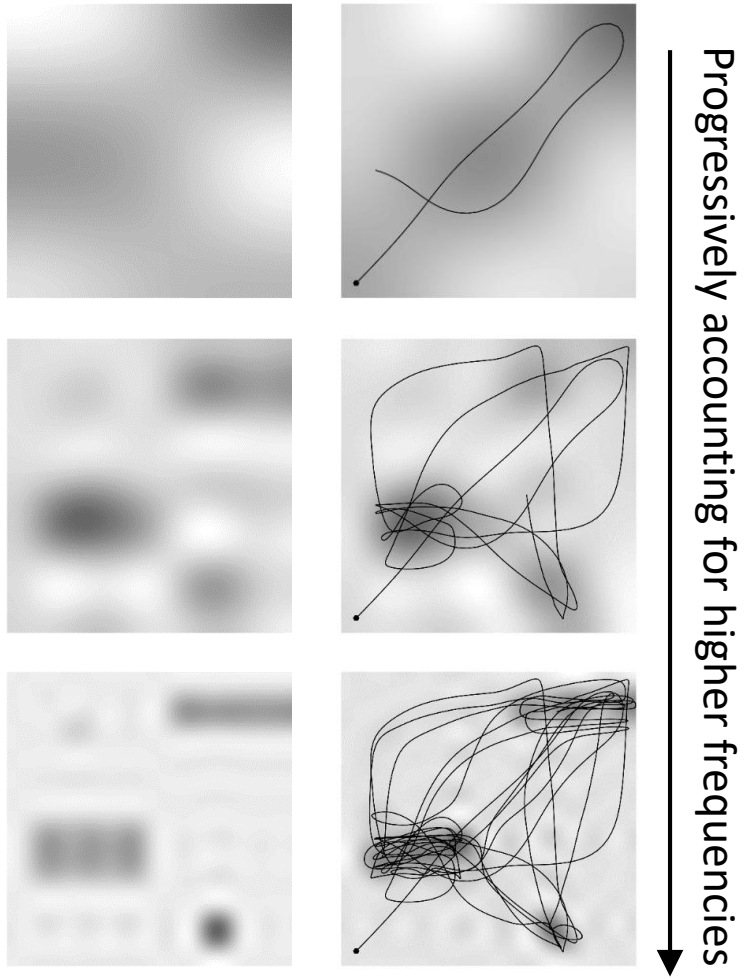






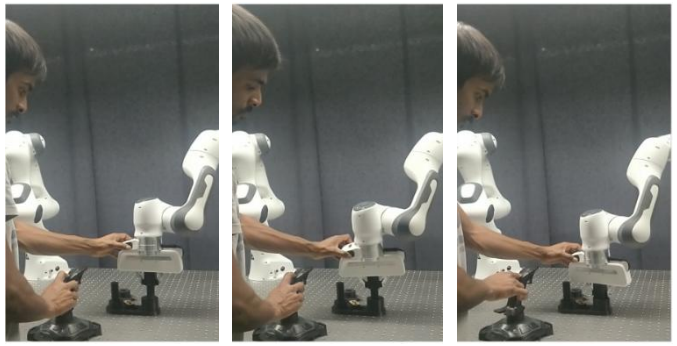


Input as probability distribution
(e.g., all flat surfaces)

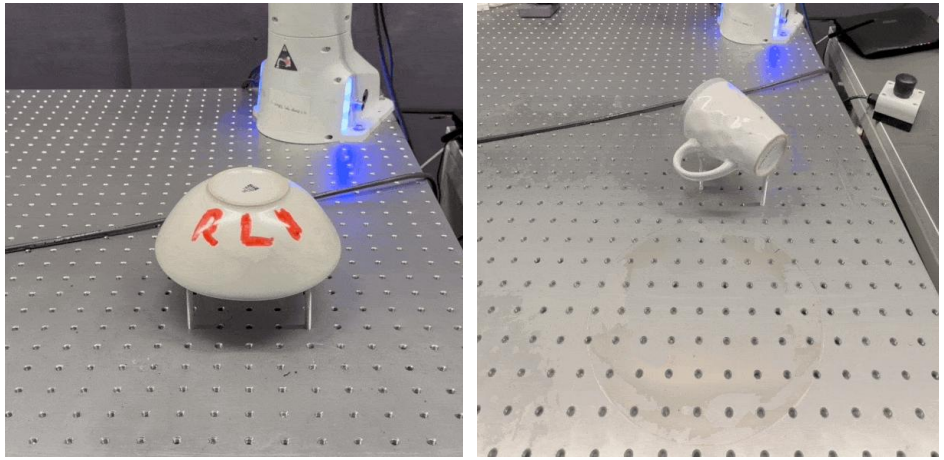


[Shetty, Silvério and Calinon, IEEE Trans. on Robotics, 2022]
https://ergodiccontrol.github.io (Tutorial at ICRA'2024)

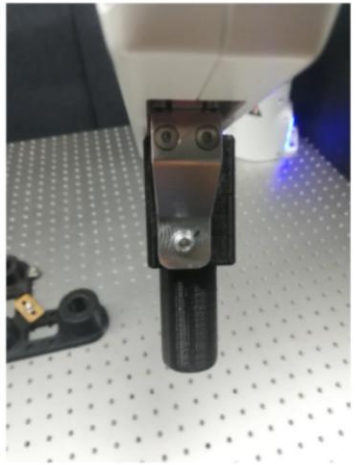
Ergodic control for manipulation with uncertain contacts



Demonstration of insertion pose variations
to provide a spatial reference distribution



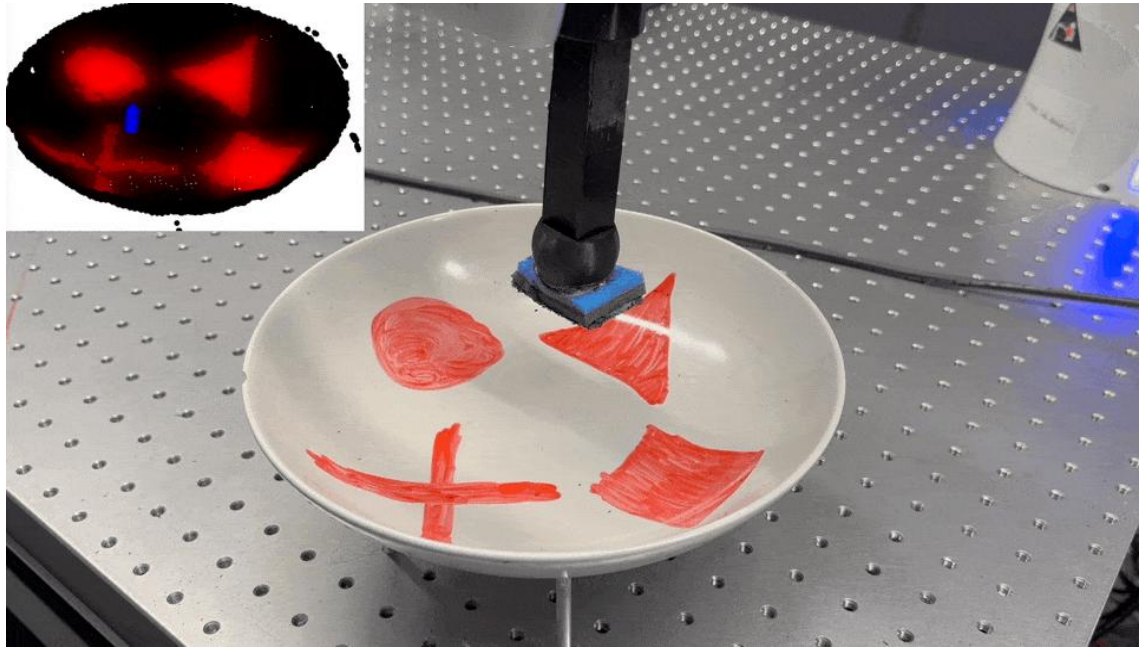
We evaluate the proposed approach using two different peg grasps:



Grasp #1



Grasp #2



[Shetty, Silvério and Calinon, IEEE Trans. on Robotics, 2022]

[Bilaloglu, Löw and Calinon, IEEE Trans. on Robotics, 2025]

Data Efficiency by Rethinking Search and Representation

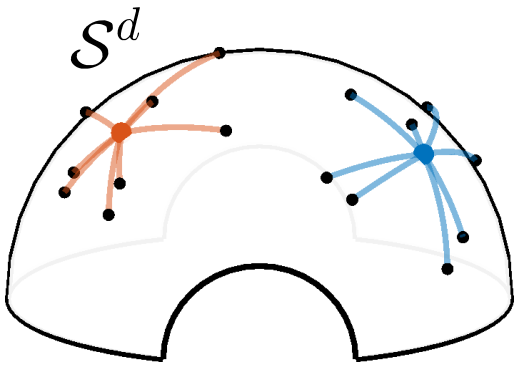
Algorithms

rethinking how to search and explore

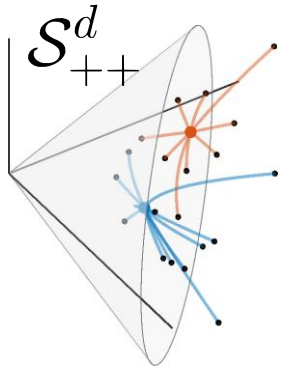
Representations

rethinking how to model robots, objects and tasks

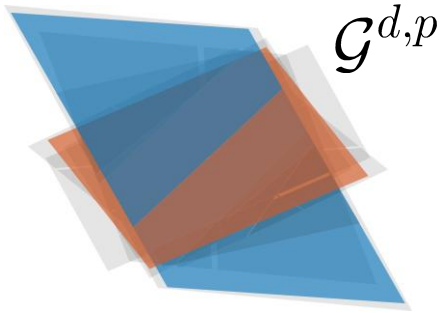
Geometric frameworks: Riemannian manifolds and Geometric Algebra



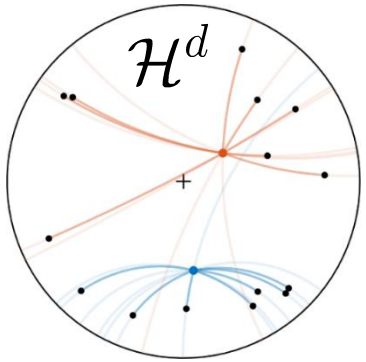
Orientations and directions
(unit quaternions, mixing coefficients, unit vectors)



SPD matrices, ellipsoids, covariances
(manipulability, stiffness, inertia, gains, graph Laplacians)



Subspaces
(planes, lines, null space projection operators, contact constraints)

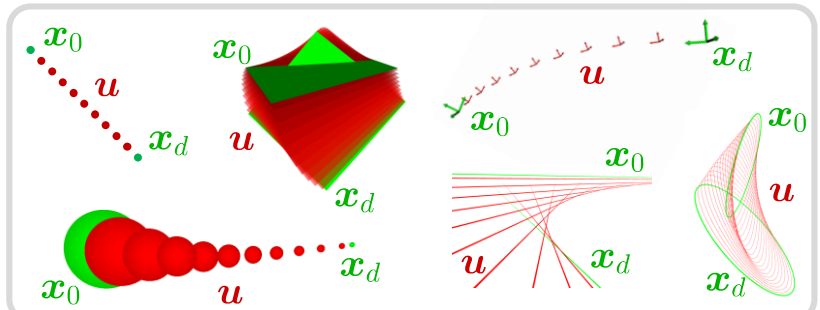


Hyperbolic embedding
(trees, hierarchies, graphs)

[Zeestraten, Havoutis, Silvério, Calinon and Caldwell, RA-L, 2017]
[Calinon, Annual Reviews, 2026]

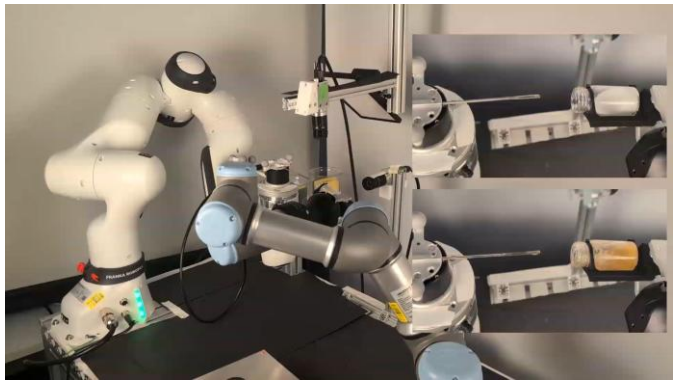
GAFRO - *Geometric Algebra For RObotics* toolbox for efficient learning and control
https://gitlab.com/gafro/

ROS



$$u = f(x_0, x_d)$$





[Löw and Calinon, IEEE Trans. on Robotics, 2023]
[Löw and Calinon, IEEE Robotics & Automation Magazine, 2025]
[Löw, Fonseca, Pruks, Deacon, Konstantinova and Calinon, Frontiers, 2026]
[Tan, Hermus, Senft, Calinon and Miéville, ChemRxiv, 2026]

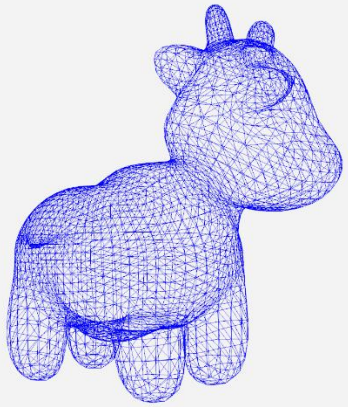


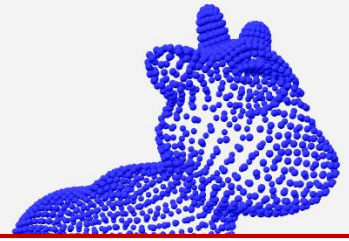
Dr Tobias Löw

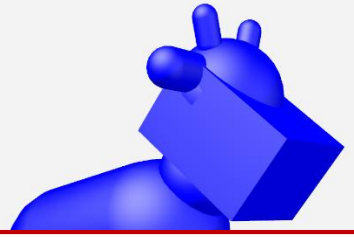
INTELLIMAN AI-powered manipulation system for advanced robotic service, manufacturing and prosthetics — Horizon Europe (2022-2025)
Sesto senso — Horizon Europe (2022-2026)
horace — SNSF (2025-2029)
ARMS — Innosuisse (2025-2026)

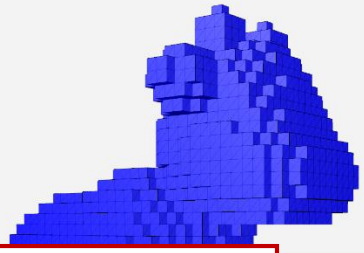
Representation of geometric shapes

Mesh **Point cloud** **Shape primitives** **Occupancy grid** **Distance function**











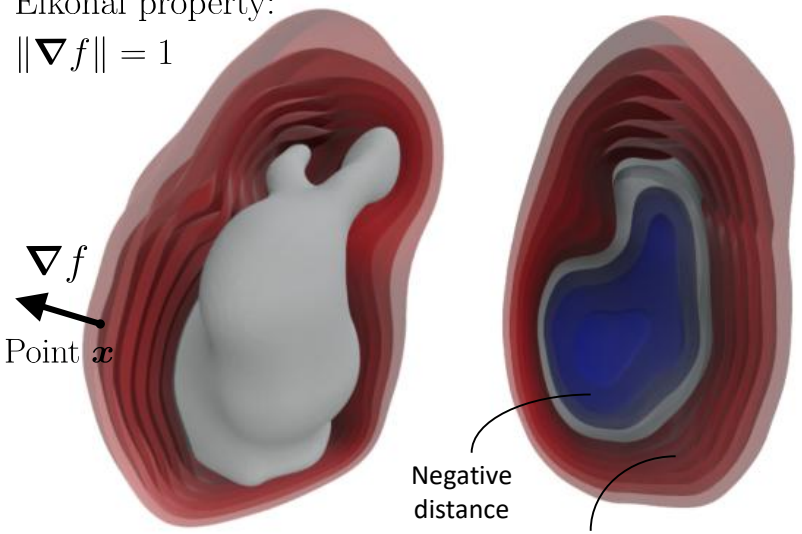


Ante Marić



Eikonal property:

$\|\nabla f\| = 1$

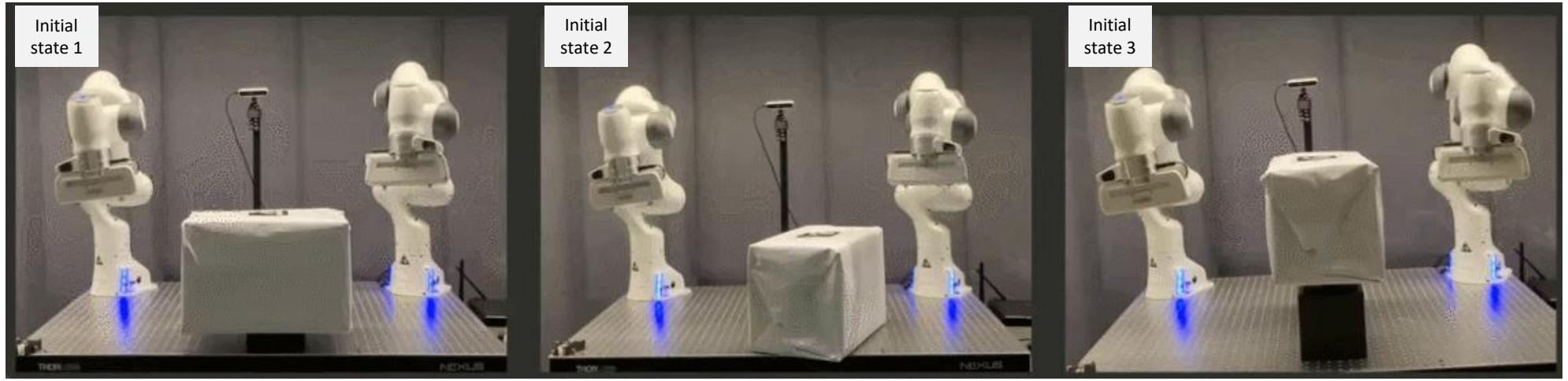


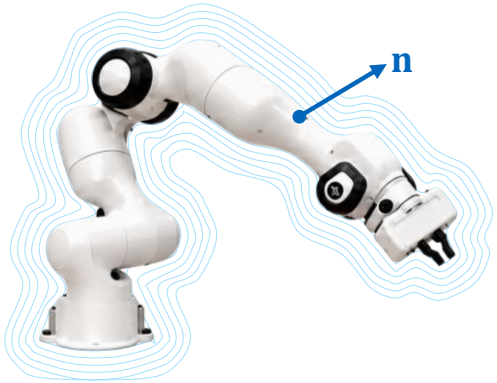
distance function $f(\boldsymbol{x})$

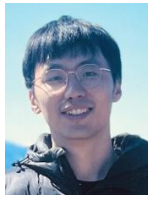
[Marić, Li and Calinon, IEEE RA-L 2024]

Signed distance fields for manipulation

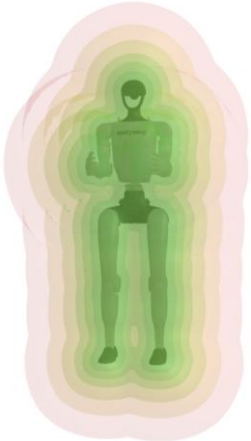
Whole-body manipulation

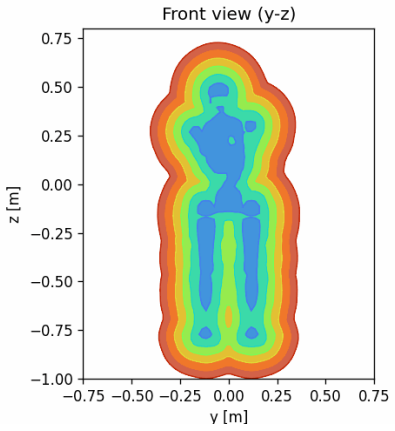


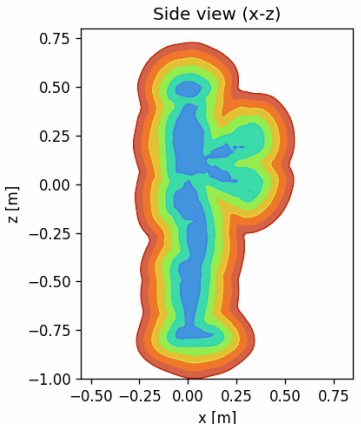




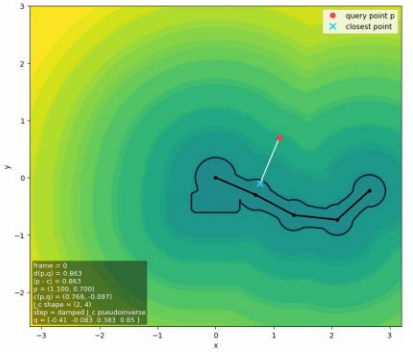
Yiming Li



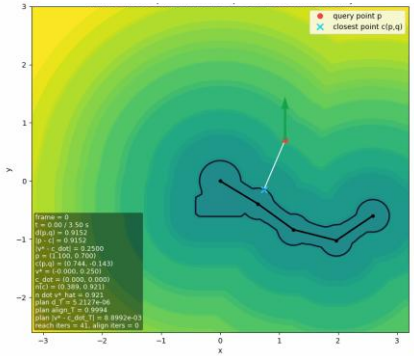




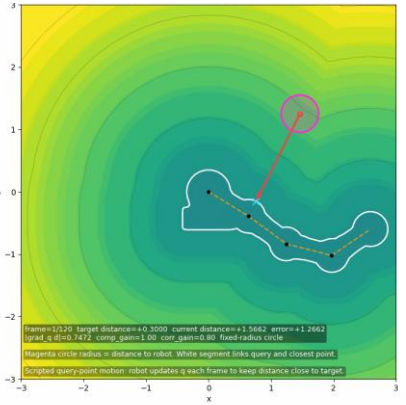
Reaching a point



Reaching a point with a given velocity



Maintaining contact with object



INTELLIMAN
AI-powered manipulation system for advanced robotic service, manufacturing and prosthetics

Horizon Europe (2022-2025)

Sesto senso

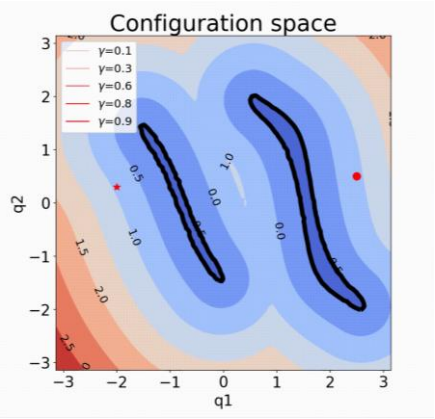
Horizon Europe (2022-2026)

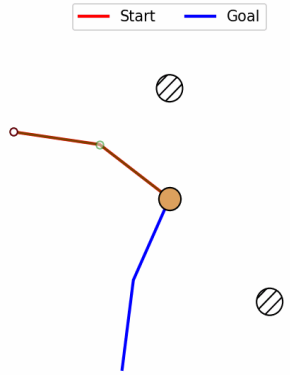
[Li, Zhang, Razmjoo and Calinon, ICRA'2024]

Signed distance fields for manipulation

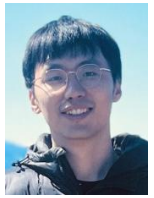
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Sesto senso
Horizon Europe (2022-2026)

Flow fields in joint angle space



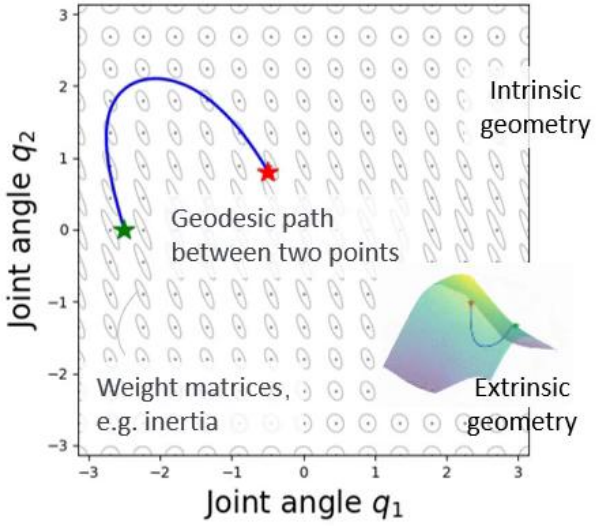




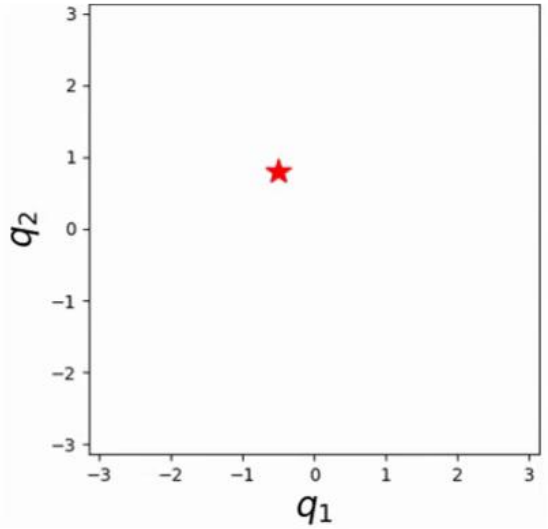


Yiming Li

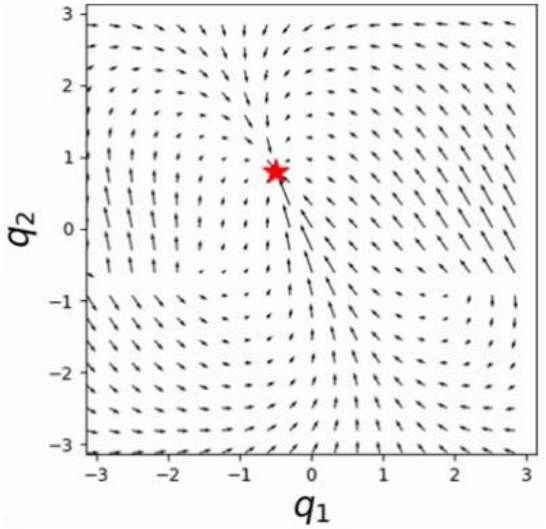
[Li, Chi, Razmjoo and Calinon, RSS'2024]
Outstanding Paper Award Finalist



Physics as
Riemannian **Metrics**



Learning Metric-Induced
Distance Fields



Geodesic Flows as
Policies

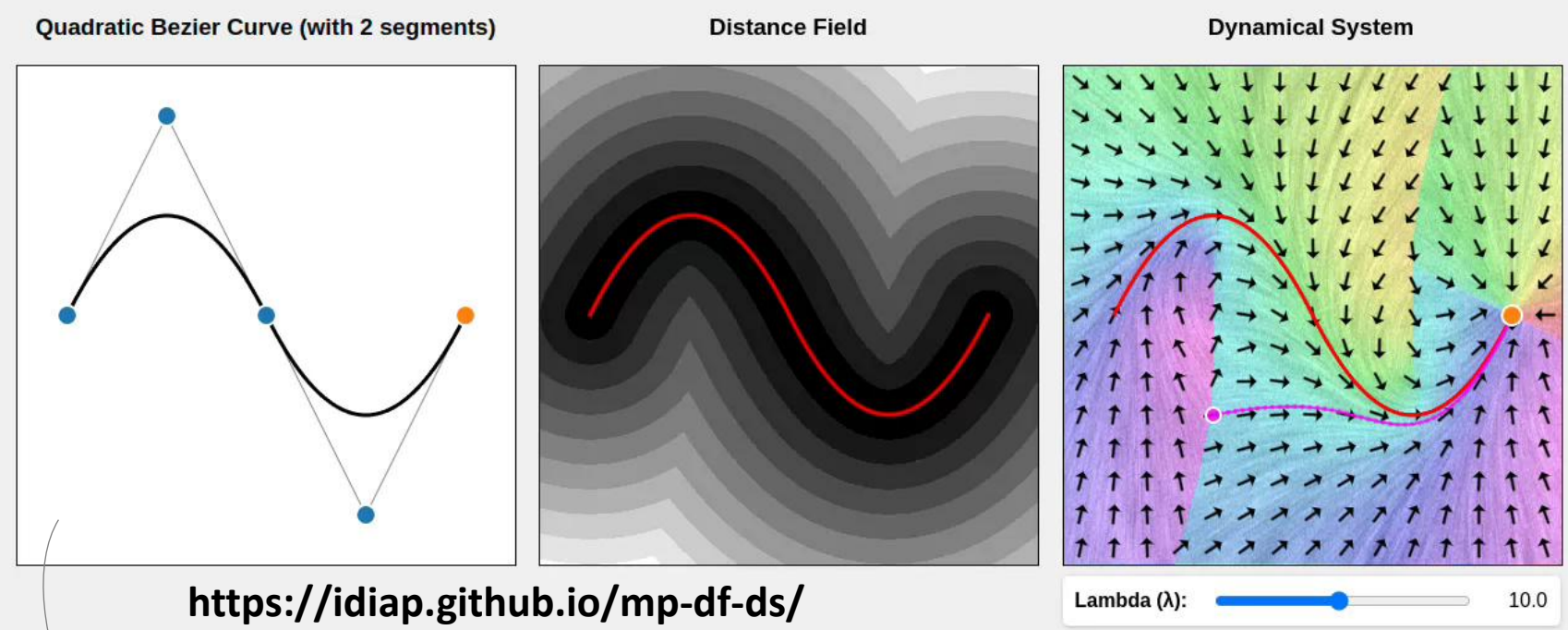
[Li, Qiu and Calinon, IJRR 2026]
Best Paper Award

Signed distance fields for manipulation

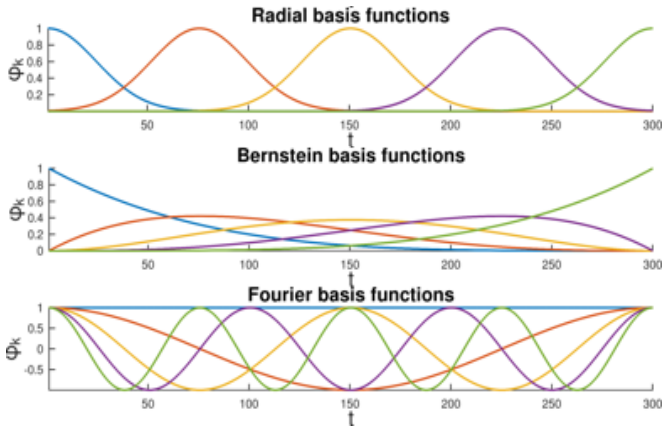
horace

SNSF (2025-2029)

From Movement Primitives to Distance Fields to Dynamical Systems



https://idiap.github.io/mp-df-ds/

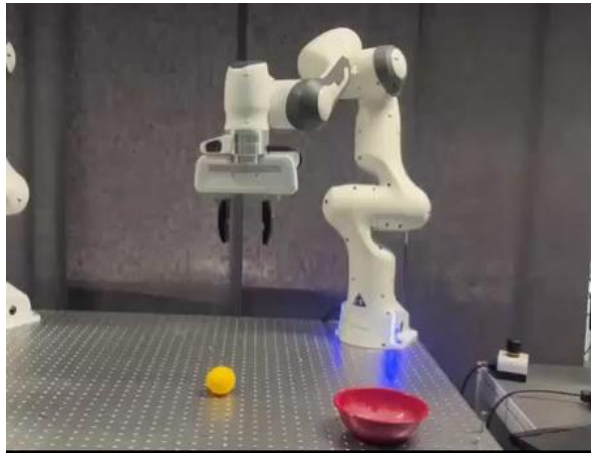


Concatenated Bézier curves can be formulated as **weighted superposition of Bernstein basis functions**, compatible with movement primitive formulations.

The approach can scale since it **learns the curve and constructs the dynamical system** instead of learning the dynamical system.

Computation time:
(laptop with basic GPU)

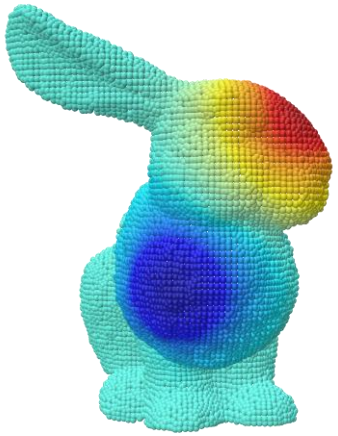
100 segments: < 10ms
100 dimensions: < 10ms



[Li and Calinon, IEEE RA-L, 2025]
[Li, Darwiche, Razmjoo, Liu, Du, Ijspeert, and Calinon, ICLR 2026]

Extension to coordinate systems

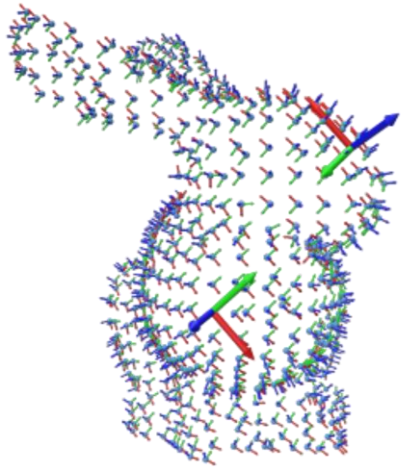
Horizon Europe (2022-2025) Horizon Europe (2022-2026)



Scalar distance field



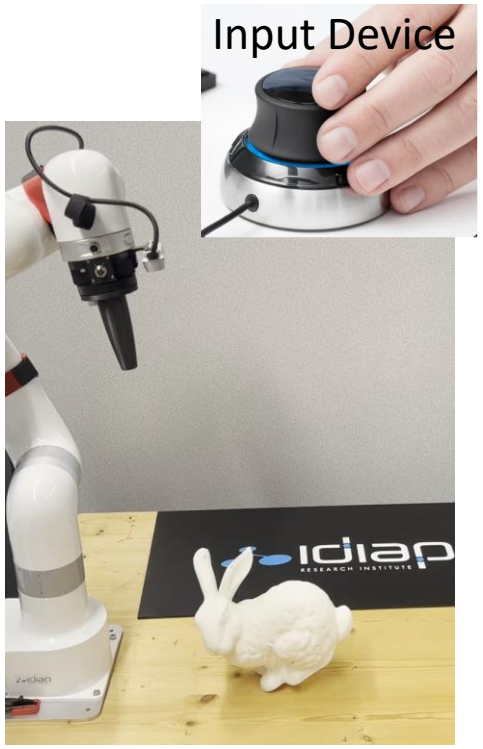
Vector field



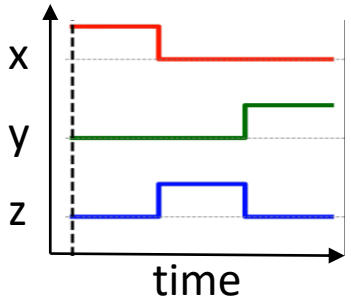
Orientation field



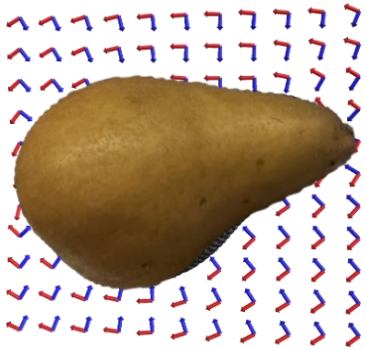
Dr Cem Bilaloglu

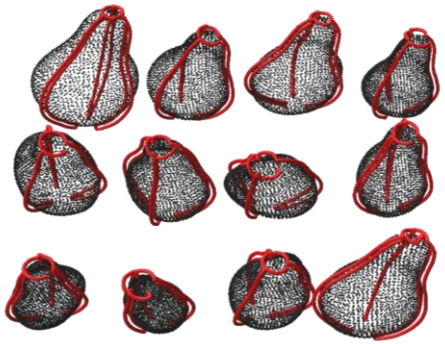






A learned policy (or teleoperation signal) is **very simple when applied to this local surface shape**!

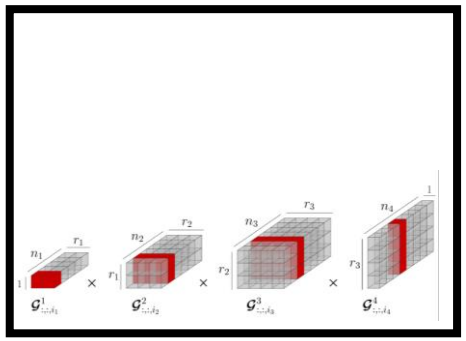




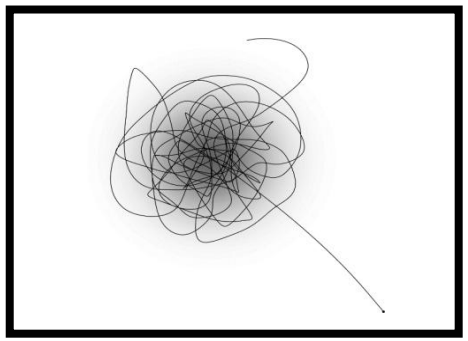
→ Object-centric representation for control, planning and learning

[Bilaloglu, Löw and Calinon, Science Robotics, 2026]

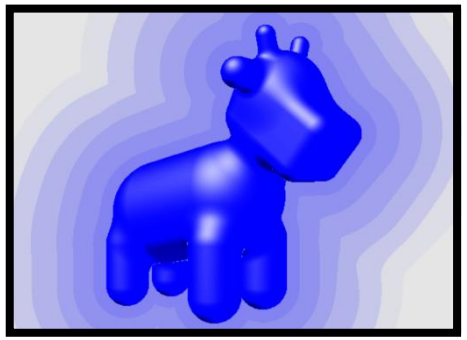
Summary



The use of **tensor trains** for skill acquisition provides a surrogate model with **frugal learning** capabilities



Ergodic control can be used for tasks with uncertainty (sensors, actuators) or tasks requiring online search, where **user inputs take the form of distributions**



Robust control and planning for manipulation can be facilitated by using **geometric** representations, incl. **implicit shapes** and **object-centric** controllers

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Slides of this presentation:

https://calinon.ch/srd2026/

Swiss National Science Foundation

Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
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Idiap RESEARCH INSTITUTE

Robot Learning & Interaction group at Idiap:

Current group members


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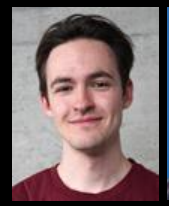

Dr Marco Frigerio

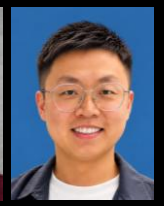

Dr Yulia Kulagina


Dr Boyang Ti

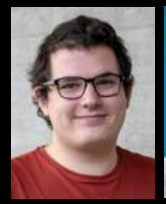

Martin Schonger


Yiming Li


Ante Marić

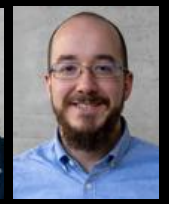

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Jérémy Maceiras


Olivier Canévet

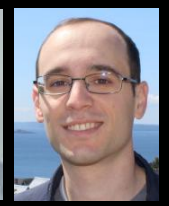

Philip Abbet

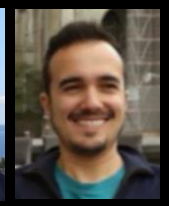

Yannick Dayer


Guillaume Clivaz

Alumni (2015-2026)

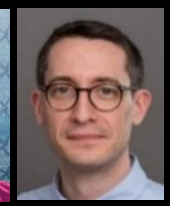

Dr Teguh Lembono

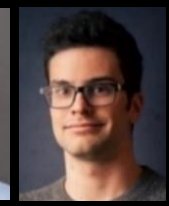

Dr Mattia Racca


Dr João Silvério

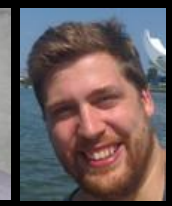

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Dr Ioannis Havoutis


Dr Emmanuel Pignat


Dr Xiao Gao


Dr Andras Kupcsik

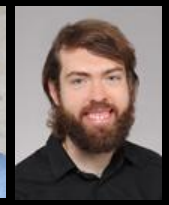

Dr Antonio Paolillo

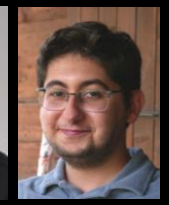

Dr Marco Ewerton

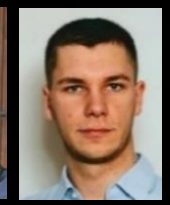

Dr Thibaut Kulak

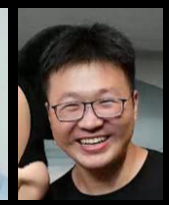

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Dr Suhan Shetty


Dr Tobias Löw


Dr Amirreza Razmjoo


Dr Julius Jankowski


Dr Teng Xue


Dr Cem Bilaloglu


Nadia Hadjmbarek

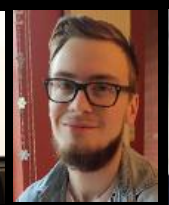

Mahdi Nobar

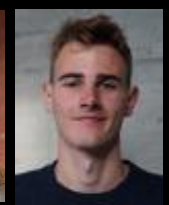

Dr Lara Brudermüller

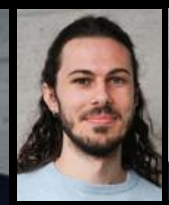

Xiaowen Jiang


Yifei Dong


Yong-Joon Thoo


Valentin Honorez

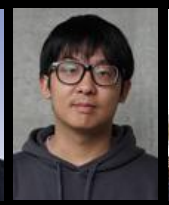

Louis Gevers


Timothé Schlüssel


Maximilien Dufau


Dr Pragna Das


Xuemin Chi


Jiacheng Qiu


Saray Bakker

RCFS Robotics Codes From Scratch.

https://rcfs.ch

SMC HEDAC

```python
# Initial robot state
param.x0 = [0.5, -0.3, 0.0, -1.8, 0.0, 1.5, 1.0]

# Number of gaussians
param.nbGaussian = 2

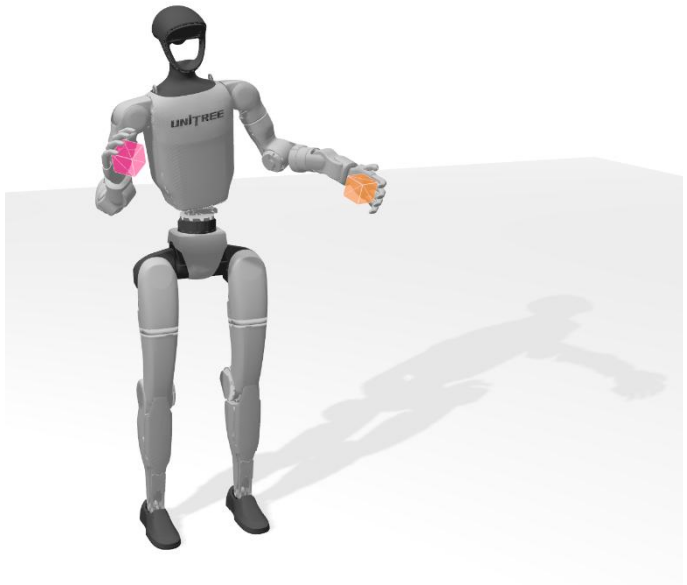

def ergodicControl(x, t, wt, param):
    # Depends on the current position only here, outputs: dphi, phix, phiy, phiz
    ang = x[:, np.newaxis] * param.rg * param.omega
    phi1 = np.cos(ang)
    dphi1 = -np.sin(ang) * np.tile(param.rg, (param.nbVar, 1)) * param.omega
    phix = phi1[0, param.xx-1].flatten()
    phiy = phi1[1, param.yy-1].flatten()
    phiz = phi1[2, param.zz-1].flatten()
    dphix = dphi1[0, param.xx-1].flatten()
    dphiy = dphi1[1, param.yy-1].flatten()
    dphiz = dphi1[2, param.zz-1].flatten()
    dphi = np.vstack([[dphix * phiy * phiz], [phix * dphiy * phiz], [phix * phiy * dphiz]])

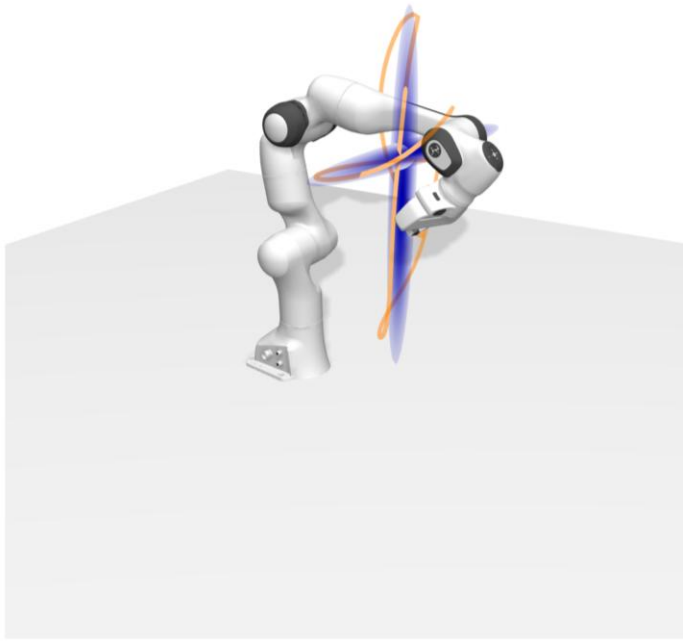
    # Depends on wt, wt starts with zeros, then updates
    wt = wt + (phix * phiy * phiz).T / (param.L**param.nbVar)

    # Controller with constrained velocity norm
    u = -dphi @ (param.Lambda * (wt/(t+1) - param.w_hat))
    u = u * param.u_max / (np.linalg.norm(u) + 0.1) # Velocity command

    # Update of the position
    x = x + u * param.dt

    return x, wt
```





</> RCFS

- 2D sandboxes
  - Forward kinematics (FK)
  - Inverse kinematics (IK)
  - Bimanual robot
  - Humanoid robot (CoM and coordination matrix)
  - Iterative linear quadratic regulator (iLQR)
  - iLQR for car
  - iLQR for bicopter
  - Impedance control
  - Jacobian
  - Ergodic control
  - Signed distance function
- 3D sandboxes
  - Inverse kinematics (IK)
  - Impedance control
  - Ergodic control
  - Physics
  - Jacobian
  - Signed distance function